

Tranquilytics: Stress-Free AI Stock Analysis for Everyone

A Senior Project Report

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by

Johann Jaramillo

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Proposal & Literature Survey

I. Introduction

Background and Significance: In today's economy, stock trading is one of the most powerful options to turn your personal wealth into something much greater than the sum of its parts. However, many working-class individuals believe stock investing to be similar to gambling. Truthfully, investing can be like gambling, but with proper research or prepared reports, said gambling can be replaced with educated guesses, often with a high chance of success. However, not all people may have the resources to quit their day job to trade full-time, nor may they have the ability or experience to do "due-diligence", the practice of researching companies before placing your trust in their growth. However, recent trends have shown that despite these drawbacks, the working class is seeing an all-time high in interest regarding the stock market¹. This specific group, a large portion of the population, is the primary target of the app Tranquilytics (Tranquility - Ease of mind; Peace + Analytics). The app will bring users the ability to make informed decisions, without the need to spend hours every day gathering intel. Furthermore, it'll avoid common pitfalls that new traders may face in noise, delayed or convoluted charts, and emotional investing. Tranquilytics will take away as many detrimental variables as it can from the stock trading process, while still holding data analytics along with the ability to share said data to users if they see the need to. The app brings power back to common, everyday people who do not seek a career in trading interested in capital gains, but it also assists those interested in the "whys" behind the market's behaviors. Tranquilytics in particular can be useful for connecting ideas by market analysts since it will include findings based on social media websites

that until recently, have not been studied such as Reddit, X (Twitter), Facebook, and more. Last of all, academic researchers may be interested in Tranquilytics due to its capacity to implement formerly exclusive practices by extensive traders, and may prove how AI provides opportunities in real-world circumstances.

Tranquilytics will employ multiple AI agents, primarily by leveraging Python's PyTorch and Pandas libraries, that utilize statistical and probability analysis as well as successful stock trading algorithms to give users stock risk ratings and/or success outcomes of placing funds into a certain stock. This data will be displayed through either a dashboard built on React/Vue or a daily statistical analysis report, which will be provided data for not only through the aforementioned tools, but also through various news sites, trading exchange discussions, and consumer sentiments on websites that often affect the market such as Yahoo Finance, StockTwits, CNN Stocks, Reddit, and many more. The goal is to learn as much as possible about a specific, high-movement stock to inform users whether it is a good time to invest or to sell. The AI will also be trained extensively using Paper Trading, the act of investing through artificial currencies, to determine whether it made good choices or not, and course-correct it based on outcomes. All of which will be presented neatly on the analysis which will include overall score, given by market sentiment, price predictions through historical data, and volatility levels. Tranquilytics will leverage three ANI agents to carry out its functionality, specializing them each to not only be experts at statistical analysis of probabilistic conditions within a market, but also rely upon and work interchangeably within themselves as brethren:

1. Technical Agent: This agent will focus on the price history and volatilities of stocks by analyzing historical data trends. This is the primary function of most software that seeks to perform automatic data gathering like Tranquilytics.
2. Sentiment Agent: The sentiment agent will use Natural Language Processing (NLP) to gather what the general opinion is on certain stocks from various social media platforms. This is an important factor that is not often taken into consideration by other analysts, as CEOs, news of deals, rumors, speculations, and discussion have all proven to rapidly affect the behavior of a stock's price.^{2,7}
3. Synthesizer Agent: The role of this final agent is to make sense of the data gathered by its siblings, weighing them differently, and producing one final confident opinion on a stock based upon the factors.

The roadmap for the production of Tranquilytics is as follows:

- Phase 1 (Weeks 1-3): Research, Literature review completion and final API selection, Progress Report #1.
- Phase 2 (Weeks 4-6): Set up the Data Pipeline, Begin developing Python scripts for automated news and stock data ingestion
- Phase 3 (Weeks 7-9): Start Developing the Technical Agent's logic, Develop and Train the Sentiment Agent via NLP and implementing the Technical Agent's logic.
- Phase 4 (Weeks 10-12): Integration, Set and Connect the Supabase backend to the React/Vue frontend dashboard.

- Phase 5: (Weeks 13 onward) Evaluation: Conduct Paper Trading Trials, Adjust and Complete Final Report and Demo.

Functional Requirements:

- 1: The system must fetch real-time and historical OHLC (Open, High, Low, Close) data using the *yfinance* or Alpha Vantage API.
- 2: The Sentiment Agent must categorize news headlines, recent market trends, and overall sentiment into "Bullish," (Likely to rise in value like a bull with horns) "Bearish," (Likely to come down in value like a bear with paws) or "Neutral" using Natural Language Processing (NLP).
- 3: The user dashboard must allow users to add and track at least 10 custom stock tickers in a watchlist.

Non-Functional Requirements:

Performance: The "Synthesis Agent" should generate a natural language summary of a stock in under 3 seconds.

Usability: The interface must be designed for "Working-Class" users, requiring no more than three clicks to view a full stock analysis report, and showing a simplified stock analysis opinion within the dashboard.

Reliability: The system should maintain 99% uptime by handling API rate limits without crashing.

II. Literature review

While Tranquilytics as a concept may seem a technology of the future, it is without a doubt that similar automated stock analytics have already been built. However, the core idea behind Tranquilytics is bringing all the attributes of trading together, while eliminating the difficulties of doing so. Apps such as TradingView provide the statistics and charts for daytrades, but charts are not fully sufficient to make an investment decision. Other apps such as TrendSpider already employ artificial intelligence as means to gather investment data on stocks, but their implementation does not include *where* the data came from, nor does it include *reasoning* behind its conclusions. Tranquilytics will not be a black box: while its inner workings won't be visible, it will state where the data was gathered from, along with giving a weight to that data. The importance of explaining the data and its origin is not simply because of interest in the topics. The ability to view these things gives users confidence that these ratings come from reasonable measurements. It also prefaces decisions before users rely upon it, so that liability is limited and frustration in case the investment is not successful may be reduced.

One key differentiating factor between Tranquilytics and competitors is that it'll maintain a competitive price point. The goal of the project is not to make a profit, but to help those struggling to break out of the lower or middle working class into upper class. If the app is truly successful, neither I nor my cooperators will need the income from it. As such, it can keep a low cost; something not very common with stock brokerages. Stock memberships, firms, or any other trading advice sources all often incur high premiums or fees, and even the free ones have been known to sneak in hidden costs once they obtain a payment method³. Of all the predecessors to Tranquilytics'

methodology, only one of the many offered a free tier ^{8,9}, and it offered limited functionality at that. Aside from that, most traders receive information through new sharing sites like Yahoo Finance, but the news on those websites come through a single contributor and are often subjective. There are also libraries for fetching market data like *yfinance* or meta's *prophet*, but again, these are not really usable since they require a proficiency in coding ability. Tranquilytics takes all of these services and websites and bundles them together into a nice, neat package through a dashboard or report, all at a low affordable price. It provides both *qualitative-quantitative* information, rather than one or the other as other services offer.

While I have not implemented my Computer Science knowledge into the open market before, I have been stock trading for the last year. Within the first month, I lost many hours of sleep because I was tirelessly studying the market whilst keeping up with my college work and part-time job. As a trader that refused to spend on subscriptions or firms because my starting point was relatively low compared to their regular clientele, I required dozens of tabs open and had to learn all the knowledge on my own, leading to great frustration, deep regret on bad choices, and eventually burnout when I stopped investing in that draining, yet successful method. This cognitive overload is not an isolated issue either; it's one that I've heard of from fellow peers and family members who've tried and failed to successfully break into the stock market.

Because of that and in only a year, I made a 25% profit on investments, which as a new trader and in a questionable market, is considered very impressive. This new venture is what inspired me to create this project, because I want others to experience the safety of passive income without the need to sacrifice physical or mental health

navigating countless websites such as I did. Tranquilytics will provide the solution by synthesizing all the data necessary on one easy-to-view dashboard.

In terms of prior work, I'm knowledgeable at Python due to CSUDH's coursework teaching it as a first language in my CSC 115 class, developing simple "Programmer's first programs" up to arrays and processing. I further reinforced it in CSC 321 Programming Languages, applying key concepts such as Recursion, Exceptions handling, and more into practice through Python. In CSC 401 Algorithms, I learned how to create algorithms for processing datasets, and I also implemented some of those algorithms in Python. In CSC 481 Software Engineering, my group and I worked with handling data and using a PostgreSQL database (supabase) in my Software Engineering class last semester where my group built an AI Tutor and shared it with the class using a dashboard developed using React. This massive project taught me the importance of version management and connecting APIs, basically backend to frontend development. Lastly, I developed and trained my own AI model from scratch in my CSC 411 Artificial Intelligence course last semester, which while I won't be doing for this project for the sake of efficiency, taught me the inner workings of Artificial Intelligence. That is, the importance of training it properly, how to combine weights to achieve desired results, learning algorithms for AI such as regression techniques, etc. All together, these foundational skills give me the courage to say that I have a good understanding of all the tools and techniques necessary for the execution of the software behind Tranquilytics.

III. Resources and References

This section will be outlining the resources the project will definitely be going to need, and if at any point they become unavailable, The project will no longer be completable

Necessary Resources

Resource	Purpose	Cost
Dell Latitude 5440 16GB RAM	Primary Workplace where my repository, IDE, and all associated libraries and applications are stored	\$0
VSCode	Primary IDE that will be used for this project. The development process will be significantly reduced if I lose access.	\$0
Supabase (PostgreSQL Database)	Cloud database that will be used to manage market data	\$0 (Free tier 500mb)
Python 3.14	The primary language utilized in the project.	\$0
Gemini Flash API or Alternative Free AI Model	Will be the base for the three models outlined in the introduction; may be swapped at some point in development	\$0 (Free tier)
Vue/React	Will be used to deploy onto a website such that the data is presented eloquently as a dashboard	\$0

Figure 1

Section II: Design and Analysis of Tranquilytics

a. Use Cases and Actors

To begin with, let us identify the actors. The Retail Trader, Market Data API, News/Social Media APIs, and lastly, the AI Synthesis Engine. This upcoming section provides a deeper look at the core interwoven interactions within Tranquilytic. The details defined below provide the step-by-step logic that's most effectively going to fulfill the system's functional requirements.

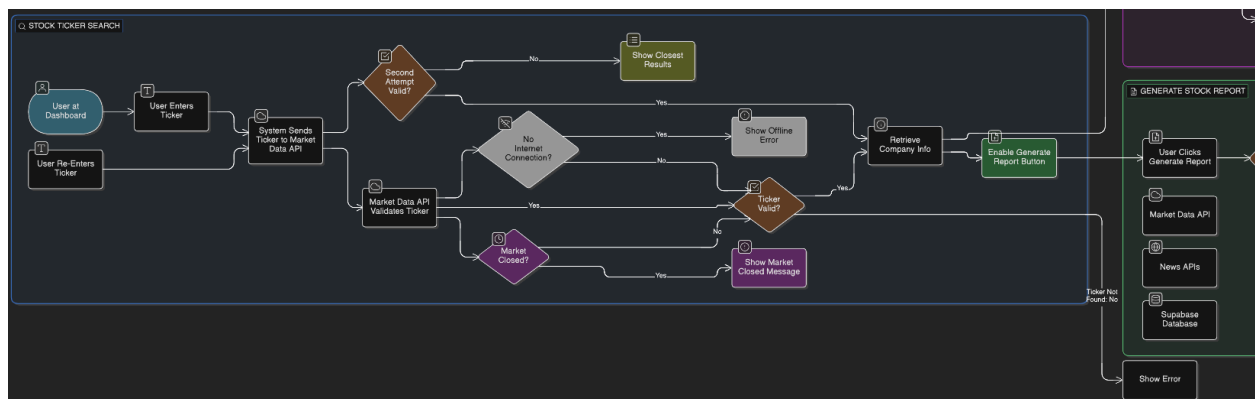


Figure 2.1

Use Case 1: Stock Ticker Searcher

Main Actor: Retail Trader

Secondary Actor: Market Data API

Pre-conditions: The user must be logged in and idling at the main dashboard

Post-conditions: The system searches for the stock and prepares the interface for a full analysis request.

Process Flow:

(Trigger) The user enters a stock symbol (e.g., "NVDA" or "MSFT") into the search bar and presses send / ENTER.

1. The system sends a request to the Market Data API to verify the ticker's validity.

2. If the ticker is valid,
 - 2.1. The system retrieves the company's full name and current trading status.
 - 2.2. Enables a button to generate a full report.
3. When the API finds no results,
 - 3.1. the system displays an error message: "Ticker not found." The system prompts the user to re-enter a ticker symbol.
 - 3.2. Display a list of "closest results" (Tickers with as many of the same characters in the same order) if the ticker cannot be found on a second entry.

Sequence Diagram for Stock Ticker Lookup

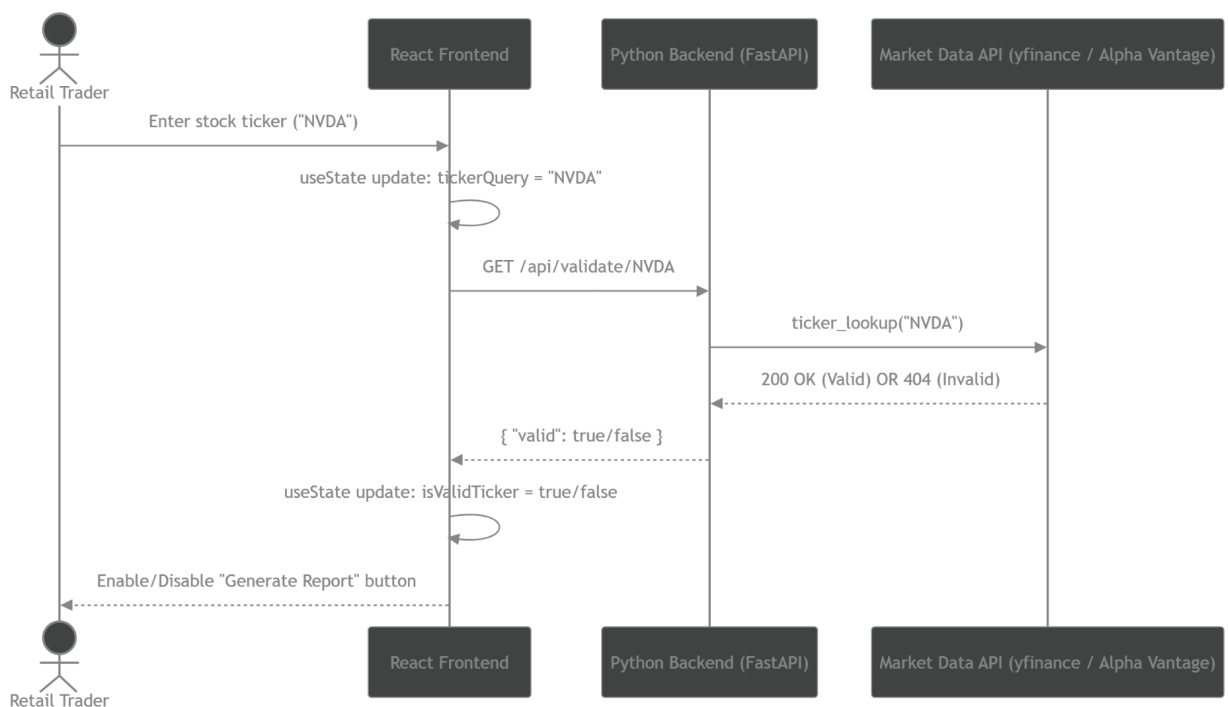
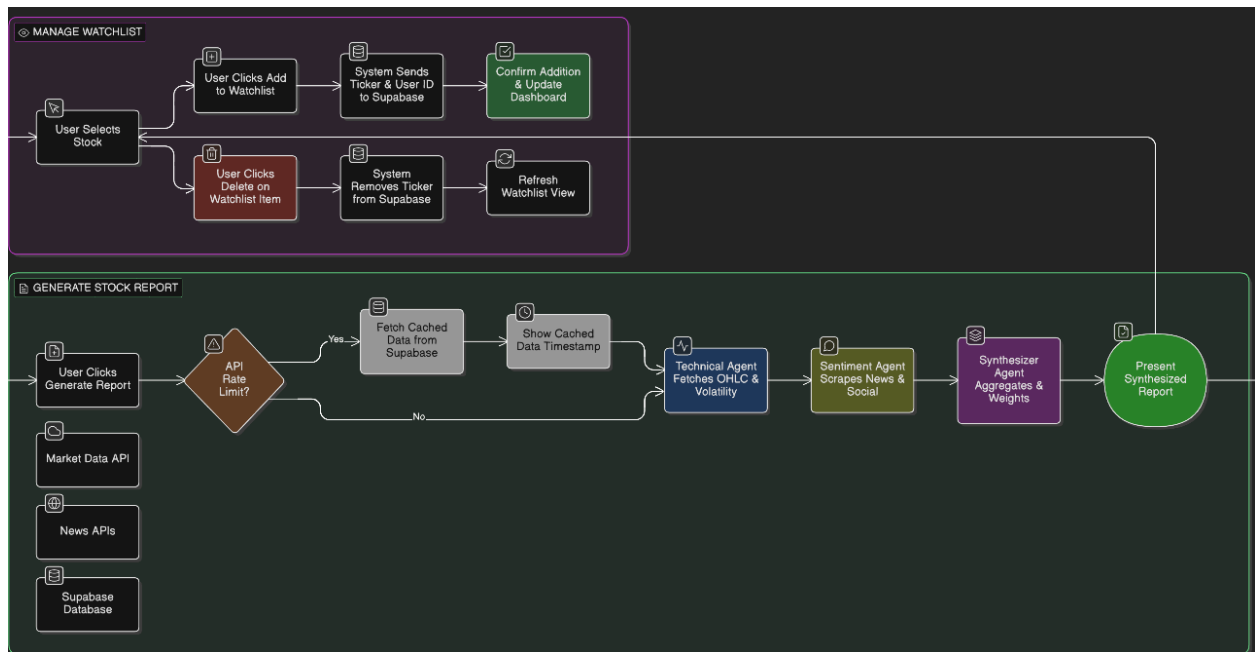


Figure 2.2

Use Case 2: Generate a Comprehensive Stock Report (Diagram Also Displays UC-3)

Figure 2.3



Main Actor: Retail Trader

Secondary Actors: AI Synthesis Engine (Technical, Sentiment, and Synthesizer Agents),

Market Data API, News APIs

Pre-conditions: A valid ticker has already been selected (UC 1.2).

Post-conditions: A synthesized analysis report is displayed to the user on the dashboard.

Process Flow:

(Trigger) The user clicks the "Generate Report" button in UC 1.2.2.

1. The Technical Agent fetches OHLC data and calculates historical volatility
2. The Sentiment Agent scrapes recent headlines and social media discussions to categorize the mood as "Bullish," "Bearish," or "Neutral"

3. The Synthesizer Agent aggregates the data gathered through previous both agents then,
 - 3.1. Applies weights to produce a final opinion.
4. The system presents the Synthesis Agent summary to the user.

Alternative: API Rate Limit

1. If any external API reaches its rate limit, the system retrieves the most recently cached data from the Supabase database to fulfill the request without crashing. When doing so, it will also display the date and time from which the data was cached.

Sequence Diagram for Generate Report

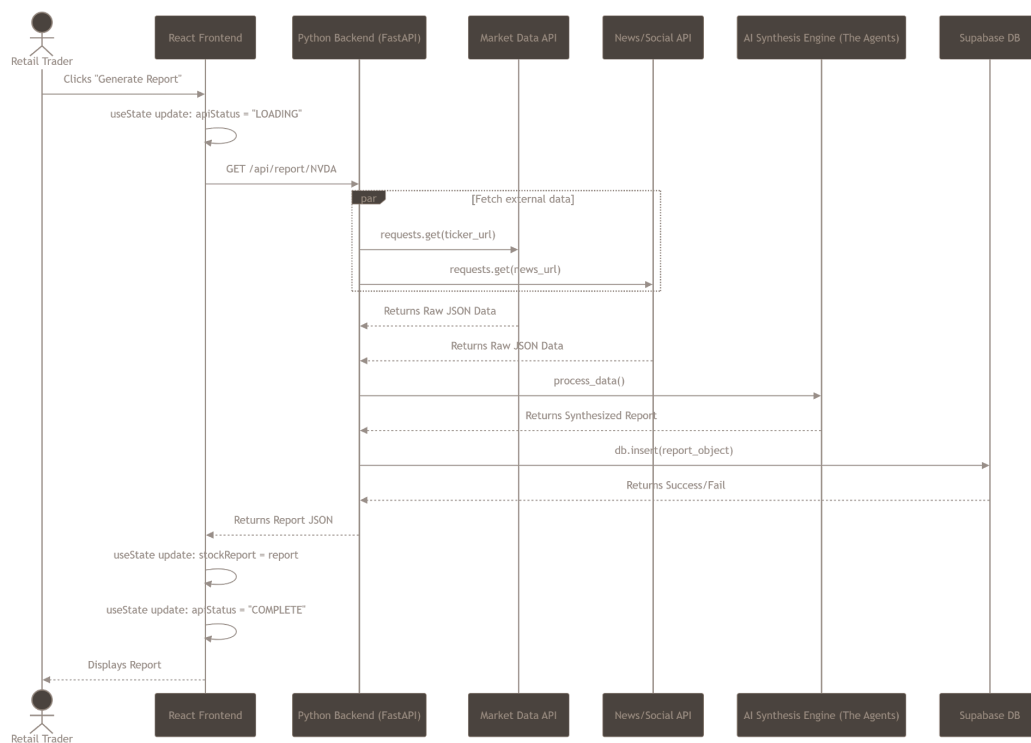


Figure 2.4

Use Case 3: Manage Watchlist

Main Actor: Retail Trader

Secondary Actors: Supabase (Database)

Pre-conditions: The user is logged into the application.

Post-conditions: The user's personal watchlist is updated and persists on the cloud.

Process Flow:

(Trigger) The user selects a stock from a search result or report (UC 1.2.2, UC 1.3.2, UC 2.4).

1. The user clicks "Add to Watchlist."
2. The system sends the ticker ID and User ID to the Supabase PostgreSQL database.
3. The system confirms the addition and updates the dashboard view to show the new item (up to 10 tickers).

Alternative Flow (Remove Ticker):

1. The user clicks the "Delete" icon next to a ticker in their watchlist.
2. The system removes the record from the database and refreshes the list.

Sequence Diagram for Manage Watchlist

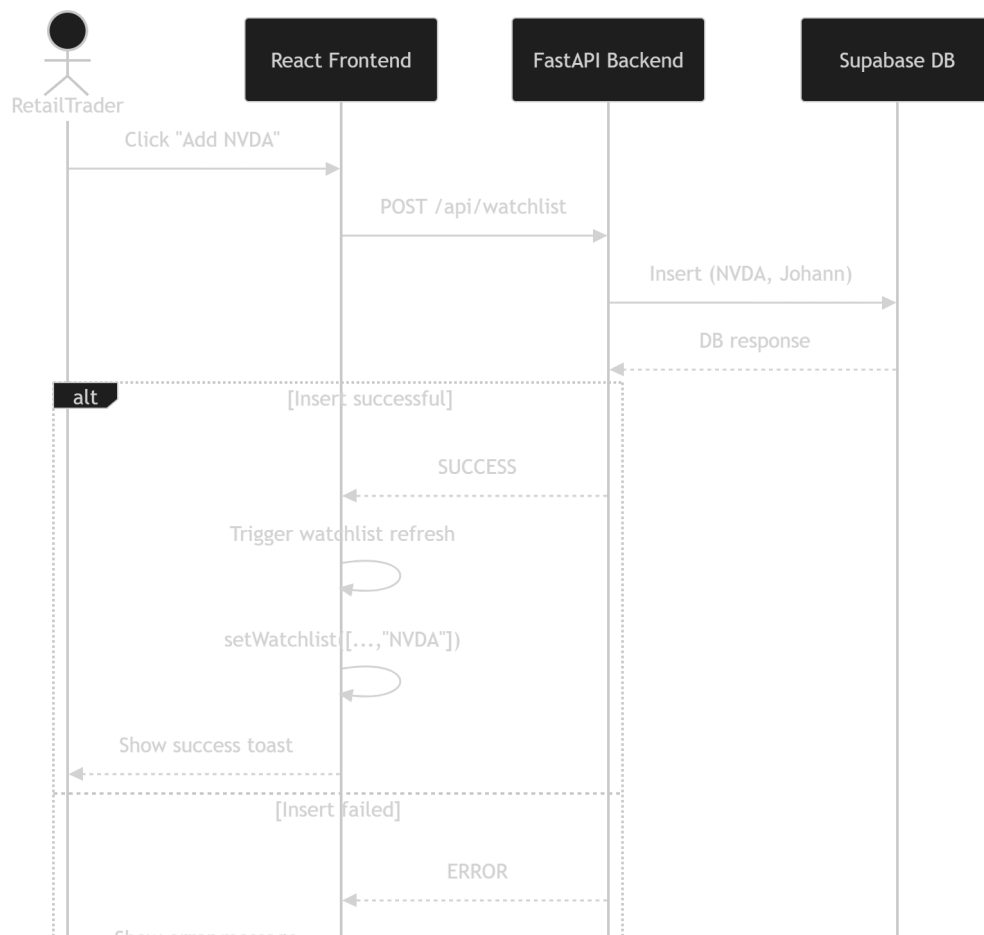
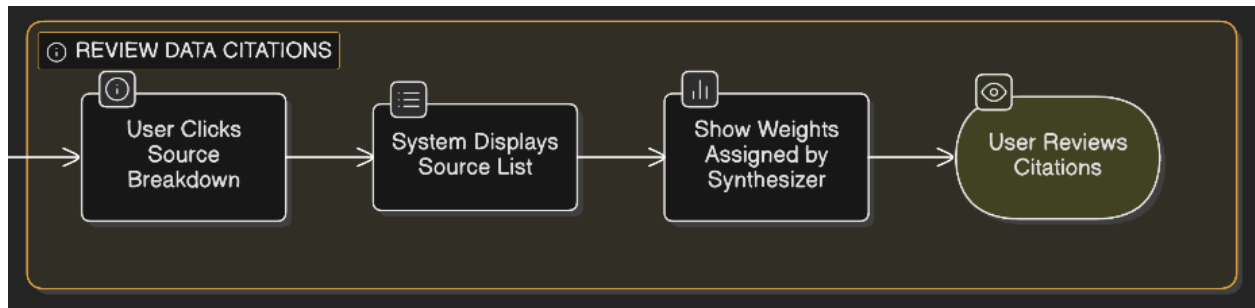


Figure 2.5

Use Case 4: Review Data Citations



Main Actor: Retail Trader

Figure 2.6

Pre-conditions: A Comprehensive Stock Report has been generated (UC 2) and the user is viewing it.

Post-conditions: The user gains transparency regarding the origin and weight of the AI's conclusions.

Process Flow

(Trigger) The user clicks on a "Source Breakdown" or "Why this rating?" dropdown.

1. The system displays a list of the specific news articles or historical data points used by the agents..
2. User reviews the weights assigned to each data point by the Synthesizer Agent

Sequence Diagram for Data Source Review F_F

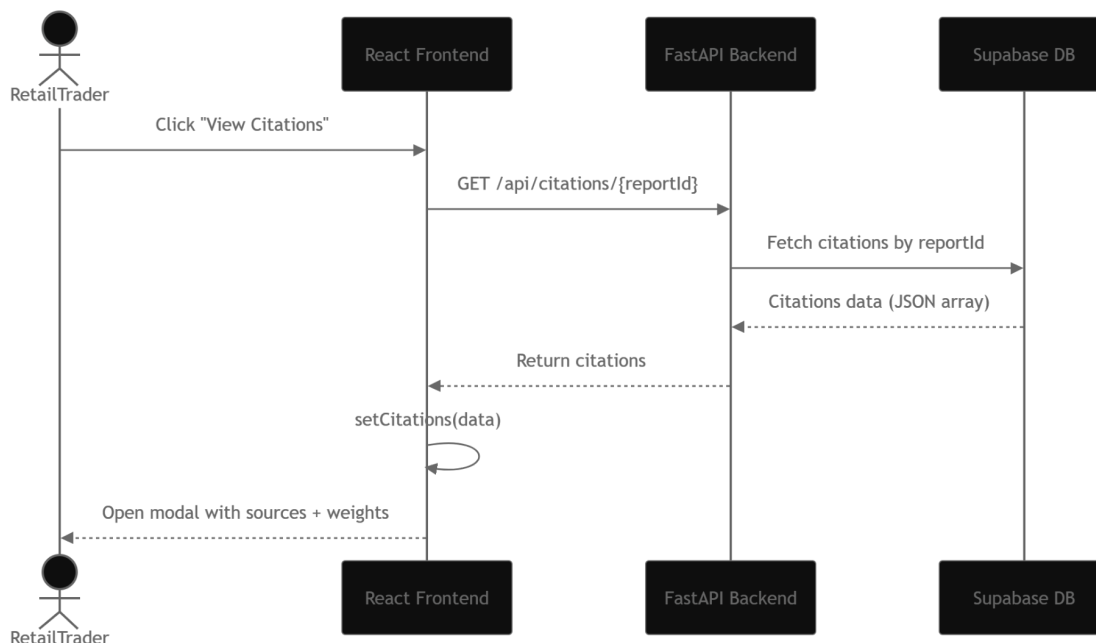


Figure 2.7

These are the use cases and sequence diagrams that Tranquilitytics will implement to satisfy the functional requirements outlined in **Section 1**.

C. Class Diagram

This is a semi-rough draft of the class diagram that represents the various classes and variables used by the AI Stock Trader

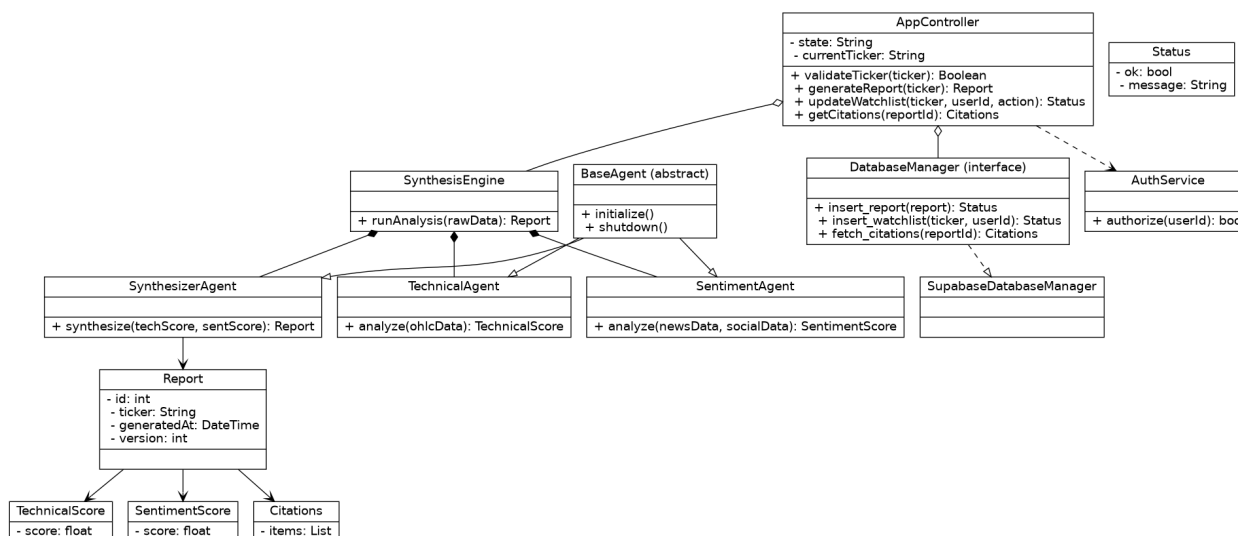
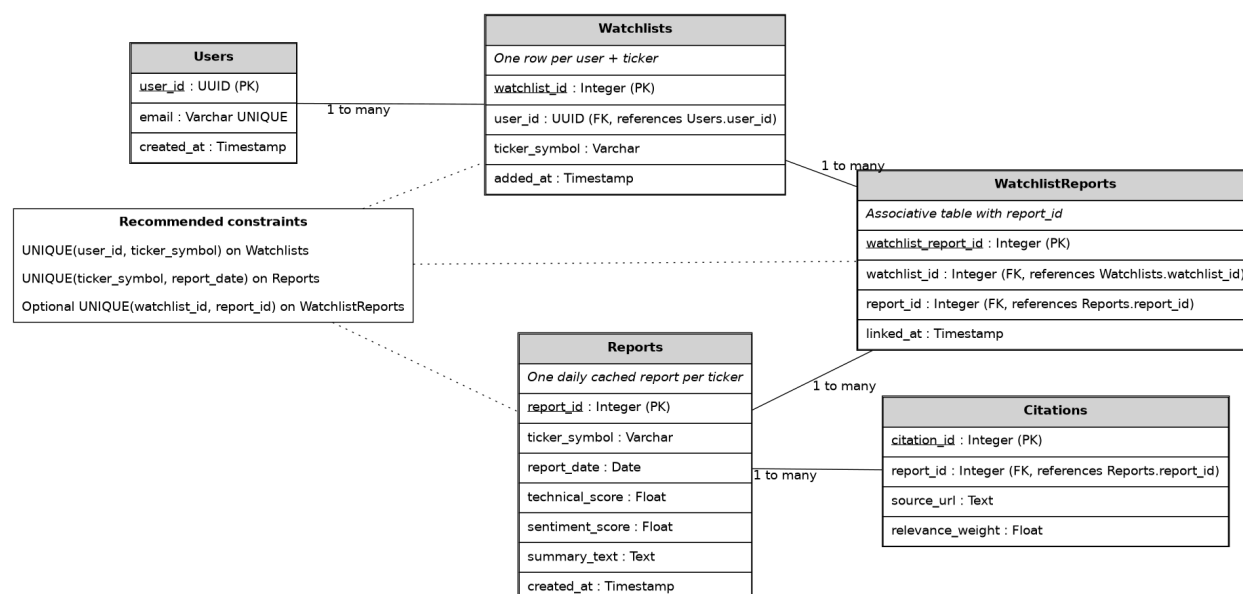


Figure 2.8

We can see that AppController is at the top of the hierarchy, and below it the SynthesisEngine utilizes each of the AI Agents derived from the abstract class to do most of the data processing, which is fed back to the AppController and relayed to the DB through the manager. Speaking of the DB, the ER Diagram is laid out in this way:

(d)

Figure 2.9



The main takeaway from the ER is the efficiency of having the entirety of reports be accessible to users, the DB saves greatly on space since new reports are only generated based on the AI's discretion. With all that said, it's time to lay the groundwork for how this software will be considered successful.

(e) Testing and SQA

The project will use the black-box method of given inputs and desired outputs to determine that it works as intended. We'll also utilize integration testing for proper communication between the React Frontend and the Python Backend.

At this time, my software is not fully ready for testing yet. Still, the test cases I'll be using are:

Test 1: Inputting a valid ticker and expecting the "analysis" button to brighten

Test 2: Inputting an invalid ticker and expect the error message and possible related searches

Test 3: Click "Generate Report Button" / Analysis, expect to receive the overall opinionated score broken down into each of the agents' conclusions

Test 4: Click "Generate Report Button" when an API is non-functional, expect fetching data from the most recent call and stating that it was not live.

Test 5: Add a ticker as a user to a watchlist, expect the ticker to persist through reloads and sessions until it is removed or unlisted.

In terms of testing the communication, I expect the general dashboard to have most user interactions take <1 second, with the report being the most heavy processing and most calls being given a leeway of <5 seconds. I also want to compare the AI's report analysis after 30 days and see what percentage of success its calls fit, with >51% not being ideal, but enough to profit consistently.

Glossary

ANI: Artificial Narrow Intelligence is a form of AI created to complete particular tasks or a narrow group of tasks with great efficiency. It operates based on set algorithms and rules. Unlike AGI, which is still a theoretical concept and would involve human-like intelligence, ANI is already in wide use. Its primary function is to handle specific duties—like translating languages or recognizing images—without being able to go beyond its programming. (Global Tech Council)

OHLC: Also known as Price Chart, Bar Chart, Open-high-low-close Charts are used as a trading tool to visualise and analyse the price changes over time for securities, currencies, stocks, bonds, commodities, etc. OHLC Charts are useful for interpreting the day-to-day sentiment of the market and forecasting any future price changes through the patterns produced. (Data Visualization Catalogue)

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